

## Case Study: Smart Voting Machine

### 1. Background

In modern SoC design, complex systems are built by integrating multiple reusable IP blocks. This case study introduces students to hierarchical RTL design and IP integration using a simple Smart Voting Machine. The design is intentionally built using basic digital building blocks such as Multiplexers, Decoders, D Flip-Flops, and Counters so that students can focus on RTL coding, simulation, integration, clocking, reset handling, and waveform analysis.

### 2. Problem Statement

Design a Smart Voting Machine that allows a user to vote for one of four candidates. The system shall select a candidate, decode the selection, register the vote, and maintain a count of total votes.

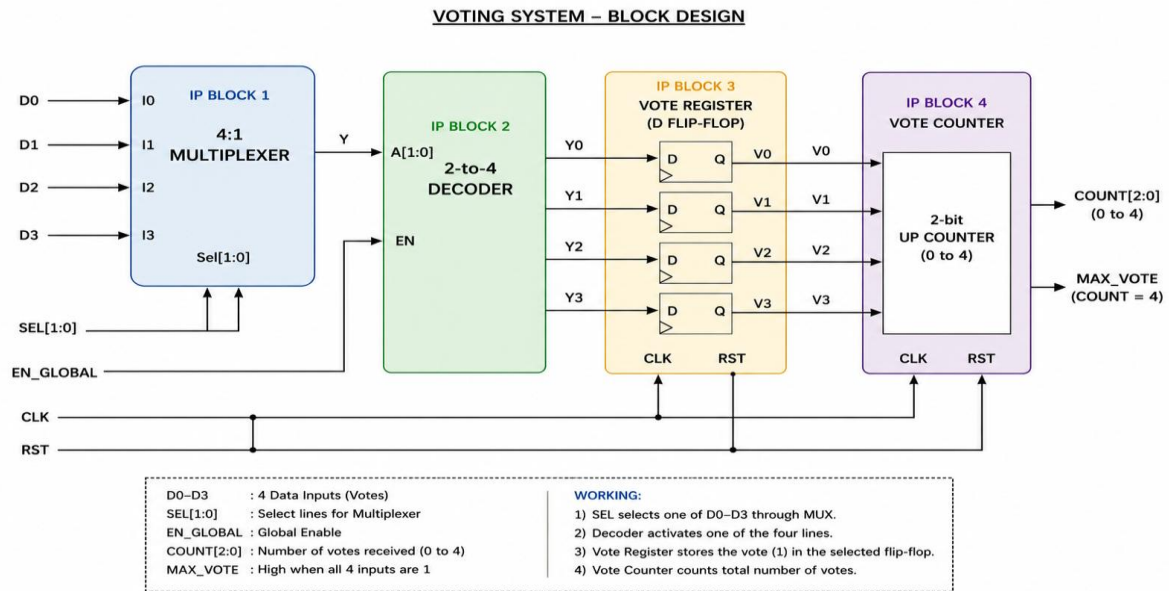
### 3. Architecture

**Block 1:** 4:1 Multiplexer

**Block 2:** 2-to-4 Decoder

**Block 3:** Vote Register (D Flip-Flop)

**Block 4:** Vote Counter



**Note :** Although these blocks are very simple these are taken for reference only .

## 4. Functional Description

| IP Block       | Purpose                               | Concept             |
|----------------|---------------------------------------|---------------------|
| 4:1 MUX        | Select one candidate from four inputs | Combinational Logic |
| 2-to-4 Decoder | Generate one-hot candidate output     | Decoder Design      |
| Vote Register  | Store vote acceptance information     | D Flip-Flop         |
| Vote Counter   | Count valid votes                     | Sequential Logic    |

## 5. Verilator Simulation Experiments

**Experiment 1:** Select Candidate A and cast a vote. Observe vote count increment.

**Experiment 2:** Select Candidate C and cast another vote. Observe decoder output and updated count.

**Experiment 3:** Apply reset and verify vote count returns to zero.

**Experiment 4:** View waveforms in GTKWave and correlate MUX, Decoder, Register, and Counter outputs.

## 6. Expected Learning Outcomes

- Understanding combinational and sequential logic
- Writing modular Verilog RTL
- Hierarchical design and IP integration
- Clock and reset handling
- Verilator simulation and GTKWave debugging
- Observation of signal behavior through waveforms